

JEE MAIN REVISION MASTER: CIRCULAR MOTION

2026 0 QUESTIONS 0% WEIGHTAGE ↓100%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 7 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Circular Motion — Standard Calculations & Problem Archetypes	6 PYQs	Quantitative Analysis
Moderate	Circular Motion — Core Principles & Governing Laws	4 PYQs	Core Principle
Standard	Circular Motion — Applied Cases, Reactions & Traps	2 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Circular Motion — Standard Calculations & Problem Archetypes	10ms^{-1} , $\text{quad } r=10\text{m}$, $\text{quad } \frac{r_2}{r_1} = \left(\frac{\omega_1}{\omega_2}\right)^2$
Circular Motion — Core Principles & Governing Laws	2ms^{-1} , $\text{quad } (x-2)^2 + y^2 = 25$
Circular Motion — Applied Cases, Reactions & Traps	$m\omega^2 = \mu mg$, $\text{quad } \frac{r_2}{r_1} = \left(\frac{\omega_1}{\omega_2}\right)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circular Motion — Standard Calculations & Problem Archetypes (Moderate Yield — 6 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Circular Motion.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The ..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An athlete runs along a circular track of diameter 80 m. The distance travelled and the magnitude of displacement of the athlete when he co..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." [Ans: N/A]

3.2 Circular Motion — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Circular Motion.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Circular Motion.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Circular Motion PYQs.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The ..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "One end of a string of length l is connected to a particle of mass m and the other to a small peg on a smooth horizontal table. If the p..." [Ans: N/A]

3.3 Circular Motion — Applied Cases, Reactions & Traps (Standard Yield — 2 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Circular Motion.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Shift] Q#3 (MCQ): "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "A coin placed on a rotating turn table just slips if it is placed at a distance of 4cm from the centre. If the angular velocity of the tur..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The speed of the particle is 2ms^{-1} , when the particle attains the lowest 'y' co-ordinate, the acceleration of the particle is (in ms^{-2})

- (A) $0.4\hat{j}$
- (B) $0.8\hat{i}$
- (C) $0.8\hat{j}$
- (D) $0.4\hat{i}$

Correct Answer: None

Solution Analysis: To determine the acceleration of a particle in uniform circular motion, we start with the equation of the circle: $(x-2)^2 + y^2 = 25$. This equation represents a circle with its center at $(2,0)$ and a radius r of 5 meters, derived from: $r^2 = 25 \Rightarrow r = 5$. In uniform circular motion, the centripetal acceleration a_c is given by the formula: $a_c = \frac{v^2}{r}$. Substituting the given speed $v = 2\text{m/s}$ and the radius $r = 5\text{m}$: $a_c = \frac{2^2}{5} = \frac{4}{5} = 0.8\text{m/s}^2$. When the particle is at the lowest y-coordinate, it is at the point $(2, -5)$, assuming

Benchmark Problem 2: KCET 2024 (Q#2)

An athlete runs along a circular track of diameter 80 m. The distance travelled and the magnitude of displacement of the athlete when he covers $\frac{3}{4}$ th of the circle is (in m)

- (A) $60\pi, 40\sqrt{2}$
- (B) $40\pi, 60\sqrt{2}$
- (C) $120\pi, 80\sqrt{2}$
- (D) $80\pi, 120\sqrt{2}$

Correct Answer: None

Solution Analysis: Given, diameter, $d = 80\text{m}$ $\therefore r = 40\text{m}$. Distance travelled after completion of $\frac{3}{4}$ revolution $= \frac{3}{4} \times 2\pi r = \frac{3}{4} \times 2 \times 40 = 60\pi$ m. Displacement $= \sqrt{r^2 + r^2} = r\sqrt{2} = 40\sqrt{2}\text{m}$